

Finite matrix levels in noncommutative L_p : trace-identity lower bounds at even exponents

Discovery packet for arXiv:0707.0427

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Status: substantial partial result, likely valid. For $p = 2m$ this packet constructs finite-dimensional maps which agree exactly through a prescribed matrix level and fail at the next one. The non-even finiteness questions remain open.

1 Source question and result

The source is Mikael de la Salle, *Complete isometries between subspaces of noncommutative L_p -spaces*, arXiv:0707.0427, published in the *Journal of Operator Theory* [1]. Section 3.3 (printed page 24) asks for two universal matrix levels. In the finite tracial setting, $n_{p,1}$ is the least n such that

$$\|\mathbf{1}_n \otimes \mathbf{1} + X\|_p = \|\mathbf{1}_n \otimes \mathbf{1} + (\text{id}_n \otimes u)X\|_p \quad (X \in M_n(E)) \quad (1)$$

forces the same equality at every matrix size. In the semifinite tracial setting, n_p is the least n such that every n -isometry between subspaces is completely isometric. The source proves

$$n_{2m,1} \leq m, \quad n_{2m} \leq m, \quad (2)$$

and records $n_{p,1}, n_p > 1$ for $p \neq 2$, using transposition. It leaves open whether either invariant is finite when p is not even.

The main result supplies quantitative lower bounds in the even case.

Theorem 1 (Matrix-level lower bounds). *Let $m \geq 2$ be an integer.*

- (i) *For every $1 \leq r \leq \lfloor m/2 \rfloor$ there are finite-dimensional tracial von Neumann algebras $(\mathcal{M}, \tau)$ and $(\mathcal{N}, \tilde{\tau})$, finite-dimensional subspaces $E \subset \mathcal{M}$, $F \subset \mathcal{N}$, and a linear map $u : E \rightarrow F$ satisfying (1) at size r but not at size $r + 1$.*
- (ii) *For every $1 \leq r \leq \lfloor (m - 1)/2 \rfloor$ there are such finite tracial algebras, subspaces, and a linear map $u : E \rightarrow F$ which is r -isometric but not $(r + 1)$ -isometric.*

Consequently

$$\boxed{\lfloor m/2 \rfloor + 1 \leq n_{2m,1} \leq m}, \quad \boxed{\max\{2, \lfloor m/2 \rfloor\} \leq n_{2m} \leq m}. \quad (3)$$

All counterexamples in the lower-bound statements can be taken between subspaces of finite direct sums of full matrix algebras, with faithful normalized traces and bounded, linearly independent spanning tuples.

In particular $n_{4,1} = n_4 = 2$. The new lower bound for $n_{2m,1}$ improves the transposition bound from $p = 8$ onward; the new lower bound for n_{2m} improves it from $p = 10$ onward.

These two results together prove that the hypotheses in Corollary 3.5 imply those in Theorem 0.3, and thus the result is proved. $\square$

3.3. On the necessity of taking matrices of arbitrary size. Here we discuss the necessity of taking matrices of arbitrary size in Theorem 0.3. In view of Theorem 0.3 a natural question is thus:

Let $p \in \mathbb{R}$. Consider the class $\mathcal{E}_{p,1}$ of all linear maps u between subspaces of noncommutative L_p spaces constructed on von Neumann algebras equipped with a *n.f.f.* normalized trace. Is there an integer n such that for any such $u : E \rightarrow F$, if (2) holds for all $x \in M_n(E)$, then it holds for any m and any $x \in M_m(E)$? The smallest such integer will be denoted by $n_{p,1}$.

A similar question is:

Let $p \in \mathbb{R}$. Consider the class $\mathcal{E}_p$ of all linear maps u between subspaces of noncommutative L_p spaces constructed on von Neumann algebras equipped with a normal semifinite faithful normalized trace. Is there an integer n such that for any such $u \in \mathcal{E}_p$, if u is n -isometric, then u is completely isometric? The smallest such integer will be denoted by n_p .

As was noted in the introduction, the transposition map from M_n to M_n ($n \geq 2$) shows that, except for $p = 2$, we necessarily have $n_{p,1} > 1$ and $n_p > 1$.

When $p \notin 2\mathbb{N}$, it is not clear whether $n_{p,1} < \infty$ (or $n_p < \infty$).

In the opposite direction, as announced in the introduction, when $p = 2m \in 2\mathbb{N}$, then it is not hard to prove that $n_{p,1} \leq m$ and $n_p \leq m$.

Theorem 3.6. *Let $p = 2m \in 2\mathbb{N}$. Let $(\mathcal{M}, \tau)$, $(\mathcal{N}, \tilde{\tau})$ be as in Theorem 0.3.*

Let $E \subset L_p(\mathcal{M}, \tau)$ be a subspace and $u : E \rightarrow L_p(\mathcal{N}, \tilde{\tau})$ be a linear map.

Assume that for all $x \in M_m(E)$, the following equality between the p -norms holds:

$$(41) \quad \|\forall x \in M_m(E), \quad 1_m \otimes 1_{\mathcal{M}} + x\|_{2m} = \|1_m \otimes 1_{\mathcal{N}} + (\text{id} \otimes u)(x)\|_{2m}.$$

Figure 1: The exact questions and the source upper bounds, arXiv:0707.0427, Section 3.3, printed page 24.

2 Proof intuition

At an even exponent, the $2m$ th power of an L_{2m} norm is a finite cyclic $*$ -polynomial. Fixed size r sees that polynomial only through traces of $r \times r$ coefficient matrices. The Amitsur–Levitzki identity makes the alternating polynomial in $2r$ matrices vanish identically at that size, but not at size $r + 1$. Appending one distinguished marker prevents the alternating sum from disappearing merely by cyclicity of the trace. This exhibits a direction in the level- $(r + 1)$ norm data which is invisible at level r .

The remaining issue is positivity: a formal difference of moments need not come from two tracial probability spaces. Because only finitely many moment degrees occur at $p = 2m$, finite-dimensional separation resolves this. Matrix trace evaluations span the dual of the relevant cyclic-polynomial space. Write a separating functional as a signed combination of evaluations, split it into positive and negative parts, and realize those convex combinations as traces on finite direct sums of matrix algebras. A common tagging summand makes the two spanning tuples linearly independent. This converts the trace identity into an honest linear map of L_{2m} subspaces.

For non-even p , infinitely many moment degrees occur. The same finite separation no longer matches all coefficients, which is precisely why the original non-even problem is not resolved here.

3 Preliminaries

For matrices $B_1, \dots, B_q$, write

$$\mathfrak{S}_q(B_1, \dots, B_q) = \sum_{\sigma \in S_q} \text{sgn}(\sigma) B_{\sigma(1)} \cdots B_{\sigma(q)}.$$

The Amitsur–Levitzki theorem says that $\mathfrak{S}_{2r}$ vanishes identically on $M_r(\mathbb{C})$ [2]. The following explicit evaluation also records the sharpness needed below.

Lemma 2 (Chain witness). *In M_{r+1} put*

$$B_1 = e_{11}, \quad B_2 = e_{12}, \quad B_3 = e_{22}, \quad B_4 = e_{23}, \dots, \quad B_{2r-1} = e_{rr}, \quad B_{2r} = e_{r,r+1}.$$

Then

$$\mathfrak{S}_{2r}(B_1, \dots, B_{2r}) = e_{1,r+1}.$$

In particular, with $B_0 = e_{r+1,1}$, $\text{tr}_{r+1}(\mathfrak{S}_{2r}(B_1, \dots, B_{2r})B_0) = 1/(r+1)$.

Proof. The displayed ordering multiplies to $e_{1,r+1}$. Any nonzero product of all the listed matrix units must follow the directed chain and must place each vertex loop immediately before the outgoing edge at that vertex. Thus the displayed ordering is the unique nonzero summand of the standard polynomial, and it has positive sign. $\square$

We also use an elementary finite-dimensional realization principle. A cyclic $*$ -polynomial means a noncommutative $*$ -polynomial modulo cyclic rotation of words; evaluation is always followed by a trace.

Lemma 3 (Finite tracial separation). *Let V be a finite-dimensional real space of self-adjoint cyclic $*$ -polynomials in finitely many variables. The real linear span of the functionals*

$$q \longmapsto \text{tr}_d(q(Z_1, \dots, Z_k)), \quad d \in \mathbb{N}, \quad Z_i \in M_d,$$

is V^ .*

Suppose $U \subset V$ contains the constant polynomial and $q_0 \notin U$. Then there are finite-dimensional tracial von Neumann algebras and tuples x, y such that their tracial evaluations agree on U and disagree on q_0 .

Proof. Finite matrix traces separate nonzero cyclic polynomials. One elementary way to see this is to choose a cyclic word with nonzero coefficient, place independent scalar-weighted matrix units around a directed cycle spelling that word, and read its scalar monomial in the resulting trace. Polarizing separates repeated letters and taking the adjoint handles the self-adjoint real form. Hence the displayed evaluation functionals have zero common annihilator in V , so they span V^* .

Choose $\Lambda \in V^*$ with $\Lambda|_U = 0$ and $\Lambda(q_0) \neq 0$, and write $\Lambda = \sum_j c_j \text{ev}_{Z^{(j)}}$. Since $1 \in U$, $\sum_j c_j = 0$. The positive and negative coefficient sums therefore have the same value $T > 0$. Dividing by T , each side is a convex combination of normalized matrix traces. Such a combination is the faithful normalized trace on the direct sum of the corresponding matrix algebras, with the coefficients as positive central weights. The resulting direct-sum tuples x and y have moment-functional difference Λ/T . $\square$

If linear independence of the tuples is needed, replace both by $x_i \oplus t_i$ and $y_i \oplus t_i$, use the same positive weight on the common summand, and choose linearly independent matrix units t_i . This multiplies their moment-functional difference by a nonzero scalar and makes the prescribed basis map well defined.

4 An optimal cyclic-isolation lemma

The source constructs N matrices of size $\lceil N/2 \rceil$ whose trace selects exactly the cyclic order among all permutations. The next proposition shows that its matrix size cannot be improved.

Proposition 4 (Exact isolation size). *Let $c(N)$ be the least d for which there exist $a_1, \dots, a_N \in M_d$ such that*

$$\mathrm{tr}_d(a_{\sigma(1)} \cdots a_{\sigma(N)}) = \begin{cases} 1, & \sigma \text{ is a cyclic rotation of } (1, \dots, N), \\ 0, & \text{otherwise.} \end{cases}$$

Then $c(N) = \lceil N/2 \rceil$.

Proof. The upper bound is the matrix-unit construction in Remark 1.5 of the source [1]. For the reverse bound, suppose $d < \lceil N/2 \rceil$. Then $N - 2d \geq 1$. Amitsur–Levitzki gives

$$0 = \mathrm{tr}_d(\mathfrak{S}_{2d}(a_1, \dots, a_{2d})a_{2d+1} \cdots a_N).$$

In the alternating expansion, the full ordering $\sigma(1), \dots, \sigma(2d), 2d+1, \dots, N$ is a cyclic rotation of $1, \dots, N$ only for the identity permutation: its last entry is N , so a cyclic rotation must start at 1. The assumed values therefore make the same alternating sum equal to one, a contradiction. $\square$

Combining this with Lemma 2.2 and Lemma 3.2 of the source gives a useful positive statement: level r unit-perturbation equality recovers a cyclic word of degree $N \leq 2r$ whenever the source's binomial coefficient is nonzero. In particular, for non-even p it recovers every $*$ -moment of degree at most $2r$. Proposition 4 says the direct cyclic-selector method has an exact degree horizon $2r$.

5 Proof of the unit-perturbation lower bound

Fix $m \geq 2$ and r with $2r \leq m$. Work with formal variables $X_0, X_1, \dots, X_{2r}$. For $s \geq 1$ and coefficient matrices $A = (A_0, \dots, A_{2r}) \in M_s^{2r+1}$, let $P_{s,A}$ be the cyclic $*$ -polynomial characterized by

$$\tau(P_{s,A}(x)) = \left\| \mathbf{1}_s \otimes \mathbf{1} + \sum_{j=0}^{2r} A_j \otimes x_j \right\|_{2m}^{2m}. \quad (4)$$

It has degree at most $2m$. Let U_s be the real span of the self-adjoint real and imaginary parts of all $P_{s,A}$.

Give exactly one of $X_1, \dots, X_{2r}$ a star and leave all other variables, including X_0 , unstarred. Define a coefficient functional L on cyclic polynomials by alternating the coefficients of the cyclic words

$$X_{\sigma(1)}^{\varepsilon_{\sigma(1)}} \cdots X_{\sigma(2r)}^{\varepsilon_{\sigma(2r)}} X_0, \quad \sigma \in S_{2r},$$

with sign $\mathrm{sgn}(\sigma)$, and extend it to the self-adjoint real form by taking the corresponding real or imaginary part.

Every displayed word has length $k = 2r+1$ and exactly one cyclic $*$ -to-1 transition. The coefficient formula in Lemma 2.2 of the source therefore gives, up to a fixed nonzero cyclic-normalization factor,

$$L(P_{s,A}) = c_{m,r} \mathrm{tr}_s(\mathfrak{S}_{2r}(B_1, \dots, B_{2r})B_0), \quad (5)$$

where $B_j = A_j^{\varepsilon_j}$, $B_0 = A_0$, and

$$c_{m,r} = (2r+1) \binom{m}{2r+1} + 2r \binom{m}{2r}. \quad (6)$$

This is nonzero for $m \geq 2r$; at the boundary $m = 2r$ it equals $2r$.

By Amitsur–Levitzki, (5) is zero for $s = r$, so L annihilates U_r . For $s = r + 1$, choose the chain matrices from Lemma 2, take $B_0 = e_{r+1,1}$, and choose the corresponding A_j (replace the uniquely starred coefficient by its adjoint). Then (5) is nonzero. Hence this particular $P_{r+1,A}$ does not lie in U_r .

Apply Lemma 3 in the finite-dimensional cyclic-polynomial space of degree at most $2m$, with $U = U_r$ and $q_0 = P_{r+1,A}$. We obtain finite tracial tuples x, y for which all level- r quantities (4) agree, while this level- $(r + 1)$ quantity differs. Add a common tagging summand to make each tuple linearly independent, set $E = \text{span}\{x_j\}$, $F = \text{span}\{y_j\}$, and define $u(x_j) = y_j$. Equality of the $2m$ th powers is equivalent to equality of the norms, so u satisfies the desired level- r unit-perturbation equality and fails it at level $r + 1$. This proves Theorem 1(i).

6 Proof of the ordinary-isometry lower bound

Now assume $2r + 1 \leq m$. Introduce pairs of variables (U_j, V_j) for $0 \leq j \leq 2r$ and, if $h = m - (2r + 1) > 0$, filler pairs (R_ℓ, S_ℓ) for $1 \leq \ell \leq h$. For a coefficient tuple A define the homogeneous cyclic polynomial $Q_{s,A}$ by

$$\tau(Q_{s,A}(z)) = \left\| \sum_i A_i \otimes z_i \right\|_{2m}^{2m} = (\text{tr}_s \otimes \tau)((Z^*Z)^m). \quad (7)$$

Define L_0 by alternating the coefficients of the words

$$(U_{\sigma(1)}^* V_{\sigma(1)}) \cdots (U_{\sigma(2r)}^* V_{\sigma(2r)}) (U_0^* V_0) (R_1^* S_1) \cdots (R_h^* S_h).$$

Expansion of $(Z^*Z)^m$ gives, up to one fixed positive cyclic factor,

$$L_0(Q_{s,A}) = \text{tr}_s(\mathfrak{S}_{2r}(B_1, \dots, B_{2r}) B_0 C_1 \cdots C_h), \quad (8)$$

where $B_j = A_{U_j}^* A_{V_j}$, $B_0 = A_{U_0}^* A_{V_0}$, and $C_\ell = A_{R_\ell}^* A_{S_\ell}$.

Equation (8) vanishes identically at size r . At size $r + 1$, take $A_{U_j} = \mathbf{1}$, choose A_{V_j} to be the chain matrices, choose the marker product to be $e_{r+1,1}$, and take every filler product to be the identity. Lemma 2 makes (8) nonzero. Thus a level- $(r + 1)$ homogeneous norm polynomial lies outside the span of the level- r ones and the constant polynomial.

Apply Lemma 3 to that enlarged level- r span. The same positive/negative direct-sum construction and common tagging summand produce a well-defined linear map whose M_r amplification is isometric and whose M_{r+1} amplification is not. This proves Theorem 1(ii).

Taking all admissible r gives $n_{2m,1} \geq \lfloor m/2 \rfloor + 1$ and $n_{2m} \geq \lceil m/2 \rceil$. Transposition supplies $n_{2m} \geq 2$, while (2) supplies both upper bounds in (3).

7 Verification, limitations, and novelty status

The script `code/verify_trace_identity.py` performs exact integer checks for $r = 1, \dots, 4$: sampled instances of $\mathfrak{S}_{2r} = 0$ in M_r , the explicit chain identity in M_{r+1} , the marker trace, and $c_{2r,r} = 2r$. These checks audit the algebra but are not used as proof.

The construction is existential after the trace-polynomial separation step; it does not optimize the dimensions of the direct-sum algebras. The bounds do not determine either invariant except at $p = 2$ and $p = 4$. Most importantly, they do not settle finiteness for non-even p . There the

norm expansion has infinitely many moment degrees, while Lemma 3 only realizes a separator on a finite-dimensional truncation. Producing two positive tracial distributions agreeing on the entire fixed-level data space is the remaining obstruction.

A bounded novelty search on 17 August 2026 checked the run indexes, the exact arXiv id and notation $n_{p,1}$, the section title, and primary-source queries combining fixed matrix level, n -isometry, complete isometry, subspaces, and noncommutative L_p . It found the source, the author’s thesis, and general whole-space isometry classifications, but no later answer or matching trace-identity lower bound. This is bounded evidence, not a priority claim.

Human-review recommendation. High-value specialist review is recommended. The main gates are the finite-tracial-separation lemma and the bookkeeping in (5) and (8); the Amitsur–Levitzki witnesses themselves are exact and computationally audited.

References

- [1] M. de la Salle, *Complete isometries between subspaces of noncommutative L_p -spaces*, J. Operator Theory **64** (2010), 261–298; arXiv:0707.0427.
- [2] A. S. Amitsur and J. Levitzki, *Minimal identities for algebras*, Proc. Amer. Math. Soc. **1** (1950), 449–463.